

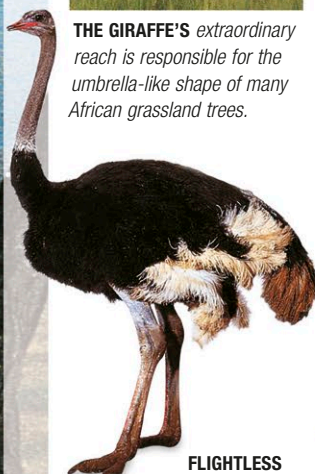
TREES AND SHRUBS



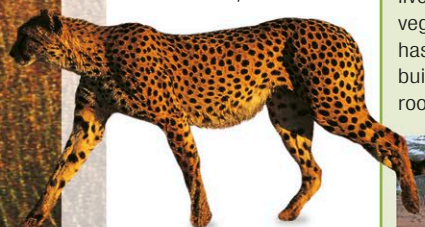
AERIAL SCAVENGERS, such as this African white-backed vulture, exploit the savanna's strong thermals, uninterrupted views, and abundance of wildlife.



THE GIRAFFE'S extraordinary reach is responsible for the umbrella-like shape of many African grassland trees.



FLIGHTLESS birds, such as the ostrich, thrive in savanna, where running is an effective means of escape.



THE CHEETAH attacks in the open, so it relies on speed, rather than stalking, to enable it to make a kill.



TERMITES carry plant matter below ground, helping to recycle vital nutrients.

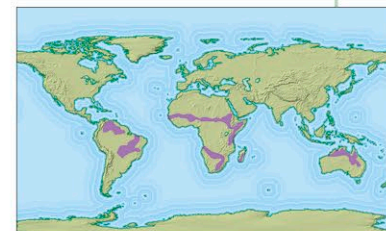
GROUND

Savanna

Savanna is tropical or subtropical grassland that contains scattered shrubs and trees. The grasslands of east Africa are a familiar example, with their diverse wildlife and distinctive vegetation (particularly the flat-topped acacia trees). Compared with temperate grassland, savanna is very variable: in some savanna habitats, trees are few and far between; in others, they form scattered thickets, merging into open woodland.

Trees have a major impact on the savanna's animal life. This is because they produce a wide variety of food, including wood, leaves, flowers, and seeds; and they also create shelter and breeding sites for animals that live off the ground. The balance between trees and grass is a delicate one that is sometimes changed by the animals themselves. For example, elephants destroy trees by pushing them over so that they can reach their leaves. However, elephants also help trees reproduce because they ingest the trees' seeds, which are then passed in their dung—an ideal medium for promoting seed growth. Browsing mammals often keep trees in check by nibbling saplings before they have had a chance to become established. Fire also helps hold back trees, and its effect is most apparent in places where trees grow close together.

Unlike temperate grassland habitats, savanna is usually warm all year round. There is often a long dry season, when most trees lose their leaves, followed by a wet, or "rainy," season, which produces a rapid burst of growth that turns the landscape green. During this wet season, plant-eating animals rarely have to contend with a shortage of food; in the dry season, the threat of starvation is never far away, and many animals travel long distances to find water and food.



SAVANNA IS FOUND mainly in Central and South America, tropical Africa, southern Asia, and northern Australia.

SOIL EROSION

At one time, natural grassland covered about two-fifths of the earth's land surface. With the spread of agriculture, much of this has been taken over for growing crops or raising livestock, leaving only small remnants with their original vegetation and wildlife. One of the side effects of this process has been a great increase in soil erosion. Grasses gradually build up soil, and hold it together with their fibrous, matlike roots. If they are removed, the soil is exposed to the wind and rain, which can strip it away. Erosion is a major problem for agriculture, because soil takes so long to form. In the short term, it cannot be replaced.



DEEP CUTS

Seen here in Africa, soil erosion can form characteristic V-shaped gullies several yards deep. Trees fall in when their roots are undermined.

HABITATS